

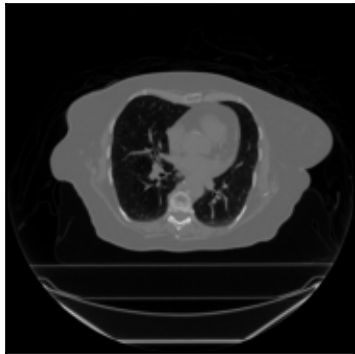
DVF QA pipeline report

DIR-Lab-case10

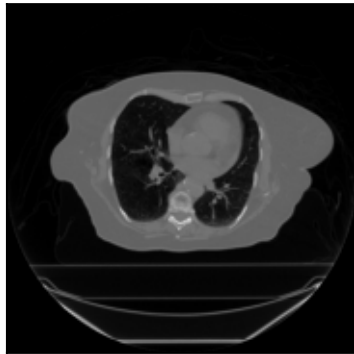
Field	Value
Case	DIR-Lab-case10
Generated	2026-05-20 09:23:37
source	Case10Pack
fps	15.0
n_cycles	4
frames_per_cycle	30
n_phases	10
ct_shape	(120, 512, 512)
spacing_xyz_mm	(0.97, 0.97, 2.5)
detector_shape	(256, 256)
pixel_spacing_mm	(1.0, 1.0)
device	cpu

4DCT phase preview

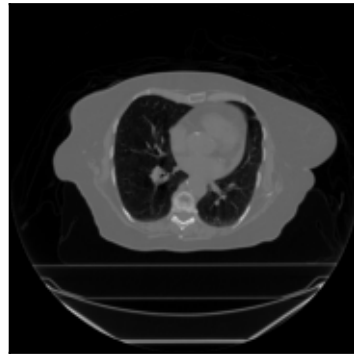
φ_0



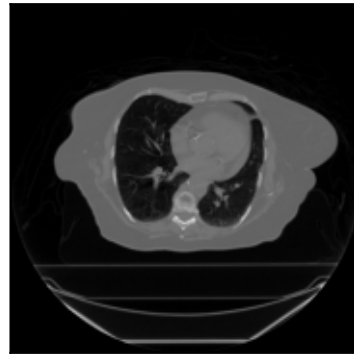
φ_1



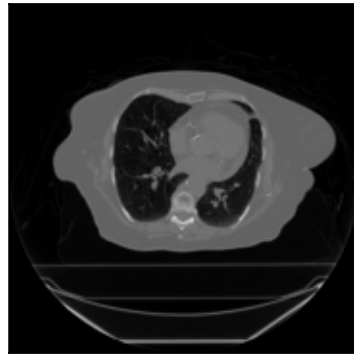
φ_2



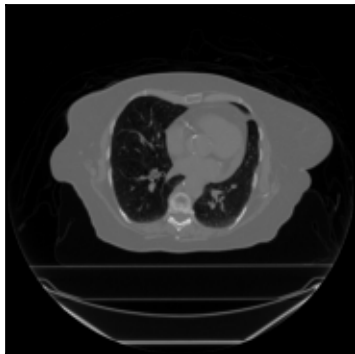
φ_3



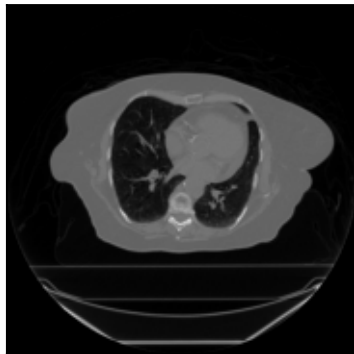
φ_4



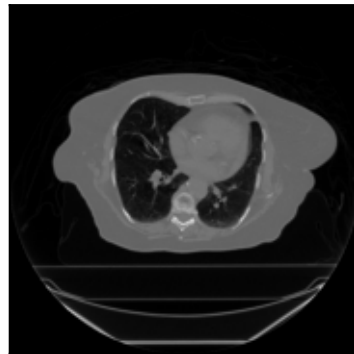
φ_5



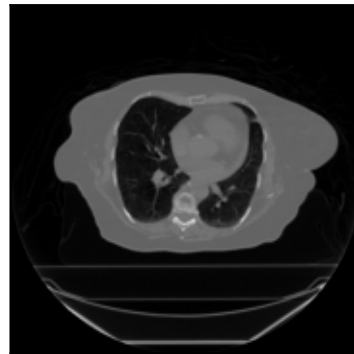
φ_6



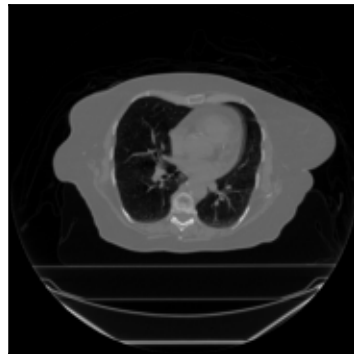
φ_7



φ_8



φ_9



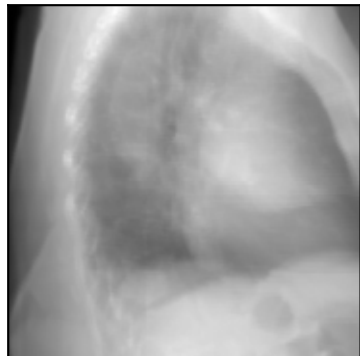
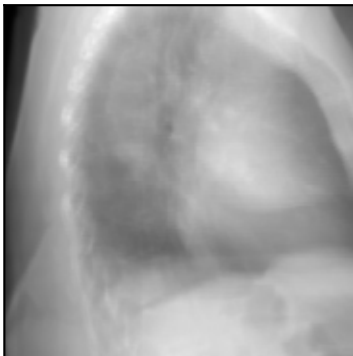
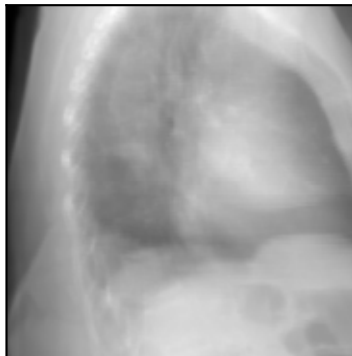
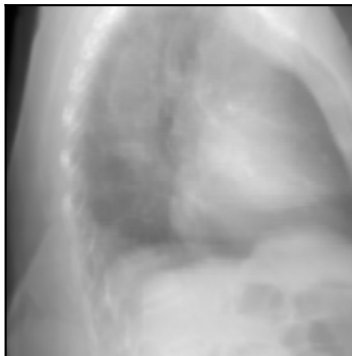
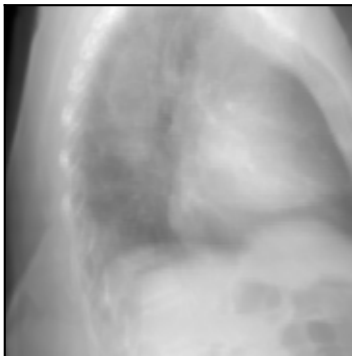
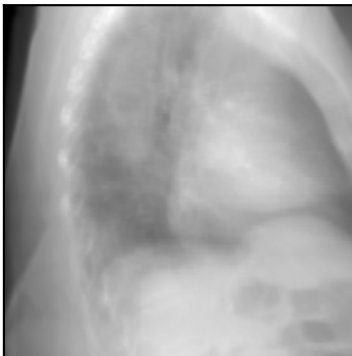
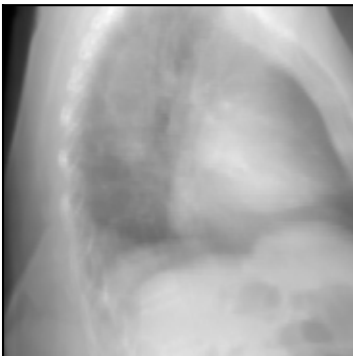
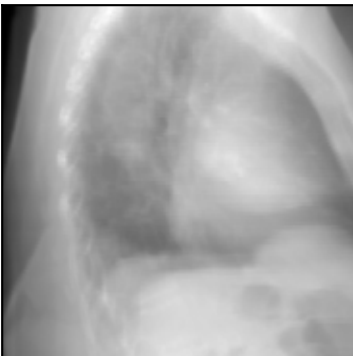
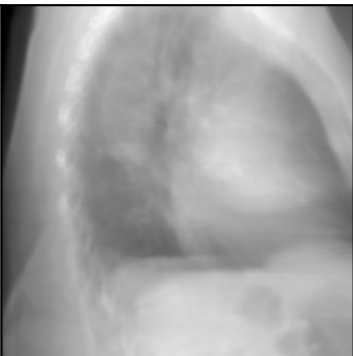
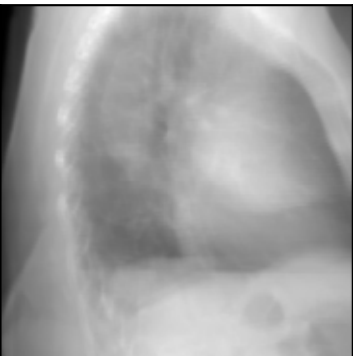
Per-phase DRRs

AP φ_0 AP $\varphi 1$ AP ϕ_2 AP $\varphi 3$ AP ϕ_4 

AP $\phi 5$



AP φ6

AP $\phi 7$ AP $\varphi 8$ AP ϕ_9 Lat ϕ_0 Lat ϕ_1 Lat ϕ_2 Lat ϕ_3 Lat ϕ_4 Lat ϕ_5 Lat ϕ_6 Lat ϕ_7 Lat ϕ_8 Lat ϕ_9 

Dynamic sequence samples

AP t=0 (0.00s)



AP t=24 (1.60s)



AP t=48 (3.20s)



AP t=72 (4.80s)



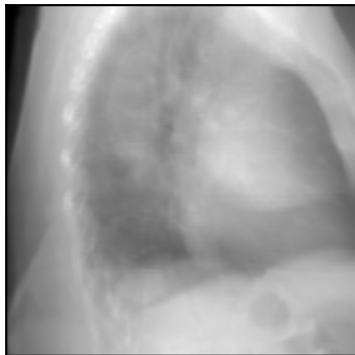
AP t=96 (6.40s)



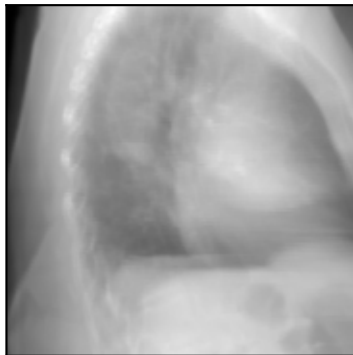
AP t=120 (8.00s)



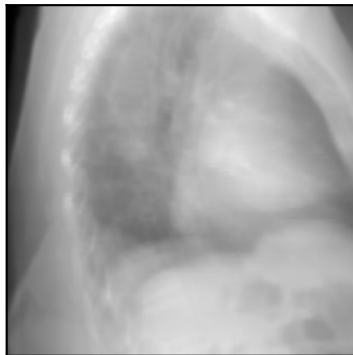
Lat t=0



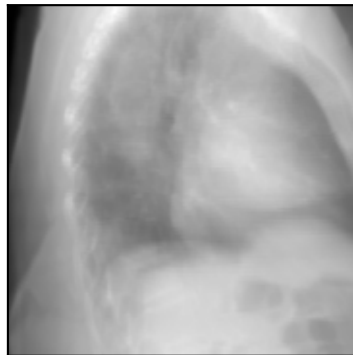
Lat t=24



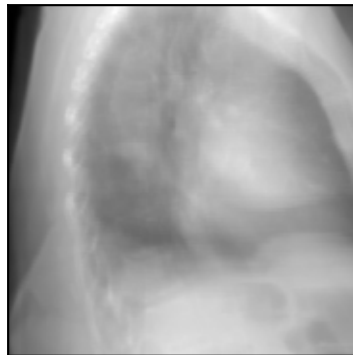
Lat t=48



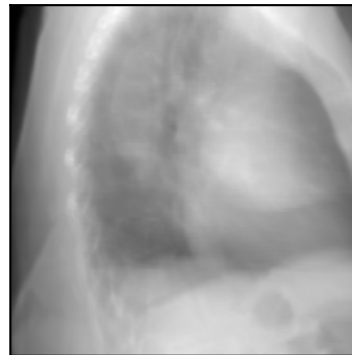
Lat t=72



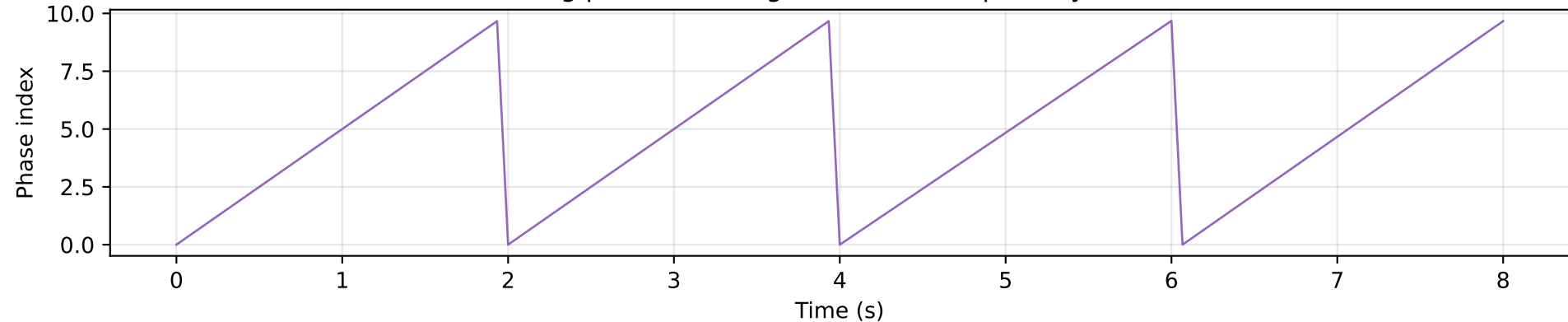
Lat t=96



Lat t=120

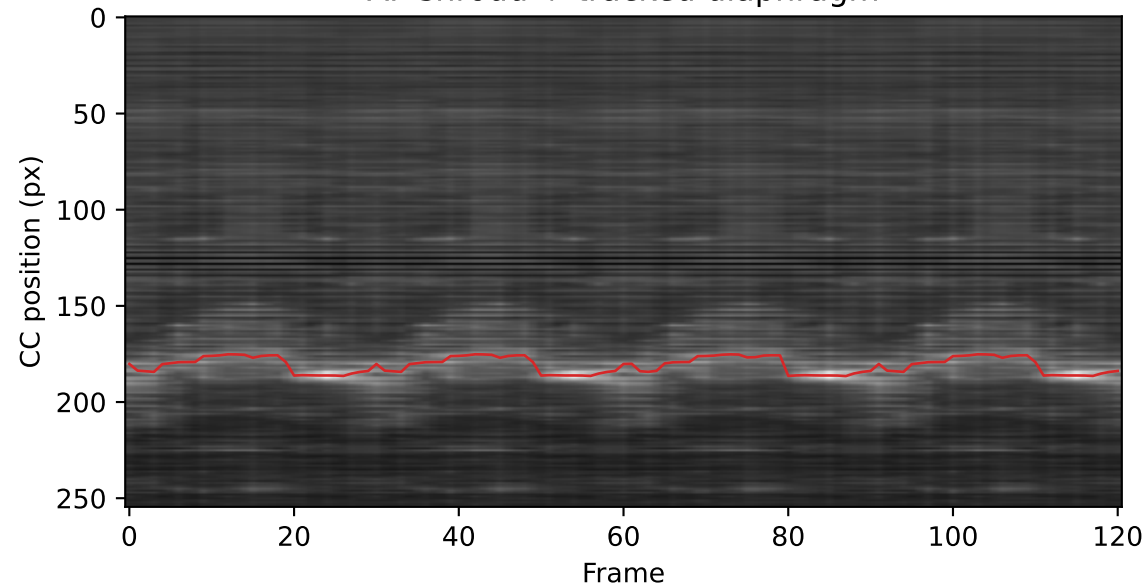


Driving phase curve
Driving phase curve (ground-truth respiratory motion)

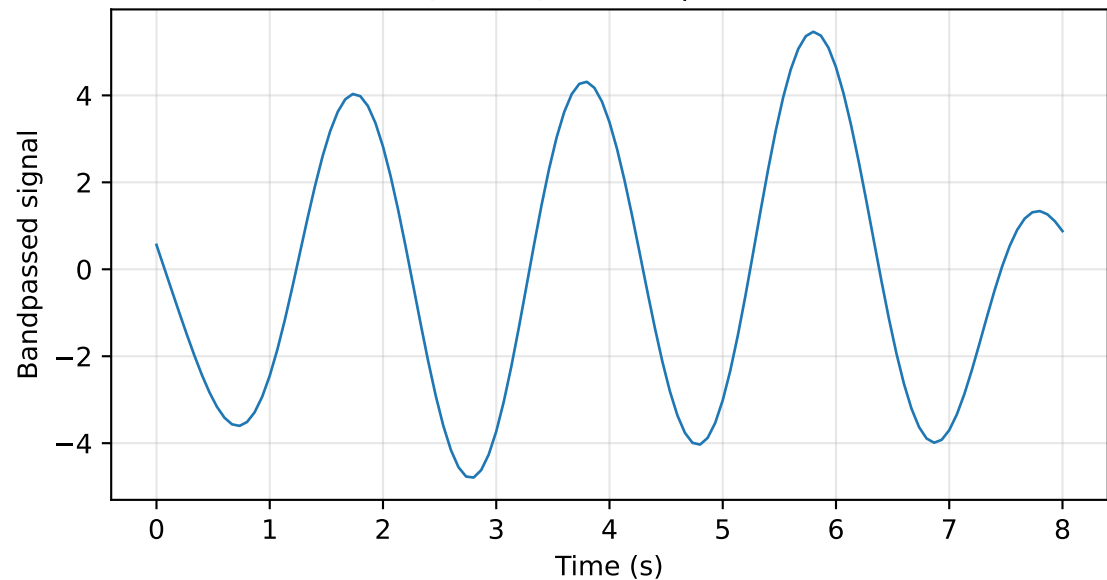


Amsterdam Shroud — AP

AP shroud + tracked diaphragm

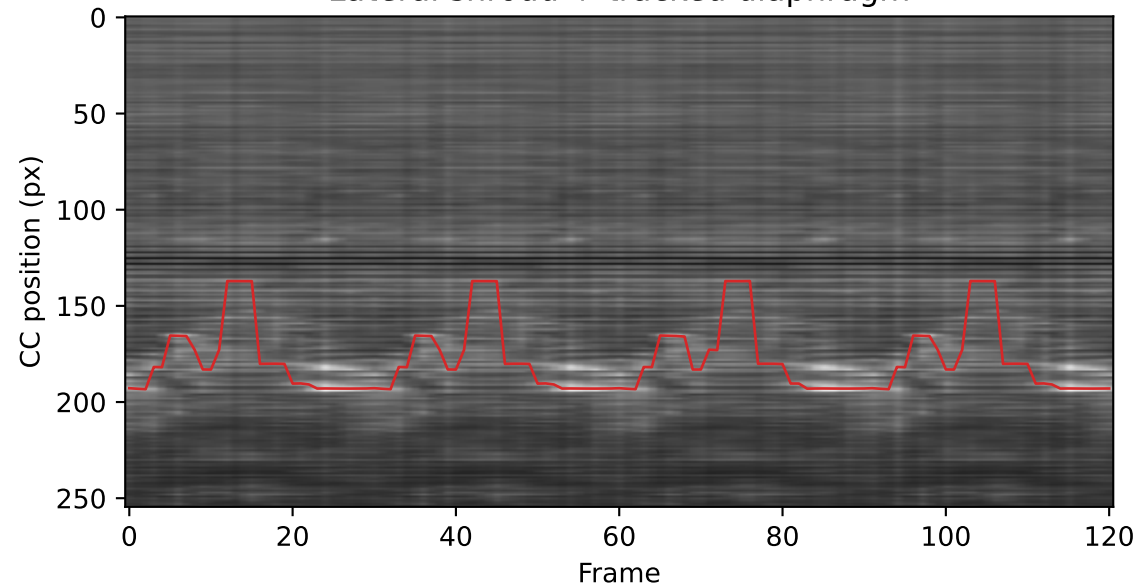


AP respiratory signal | $f=0.496$ Hz

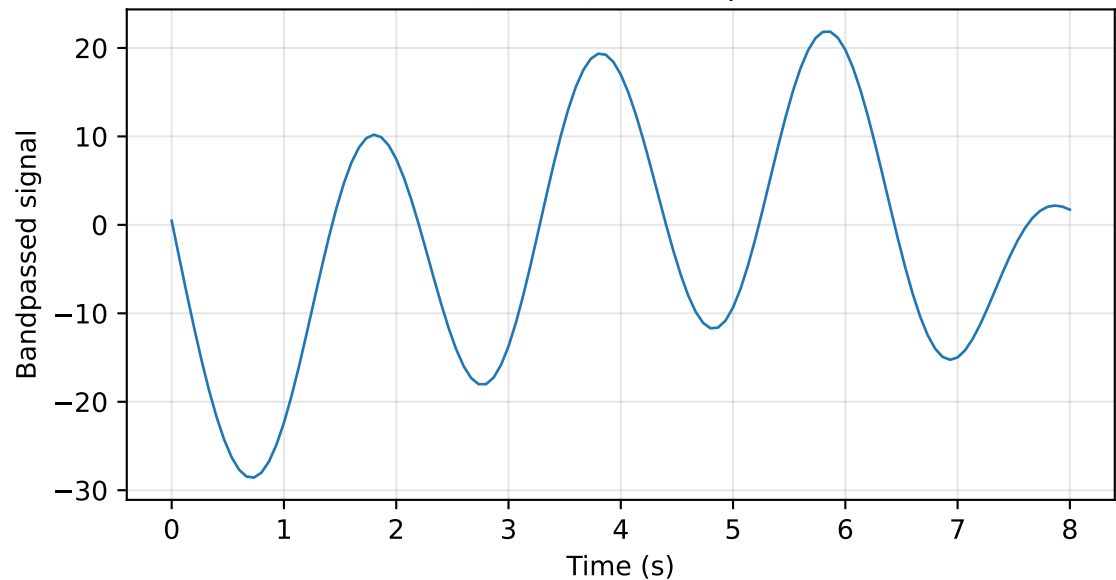


Amsterdam Shroud — Lateral

Lateral shroud + tracked diaphragm

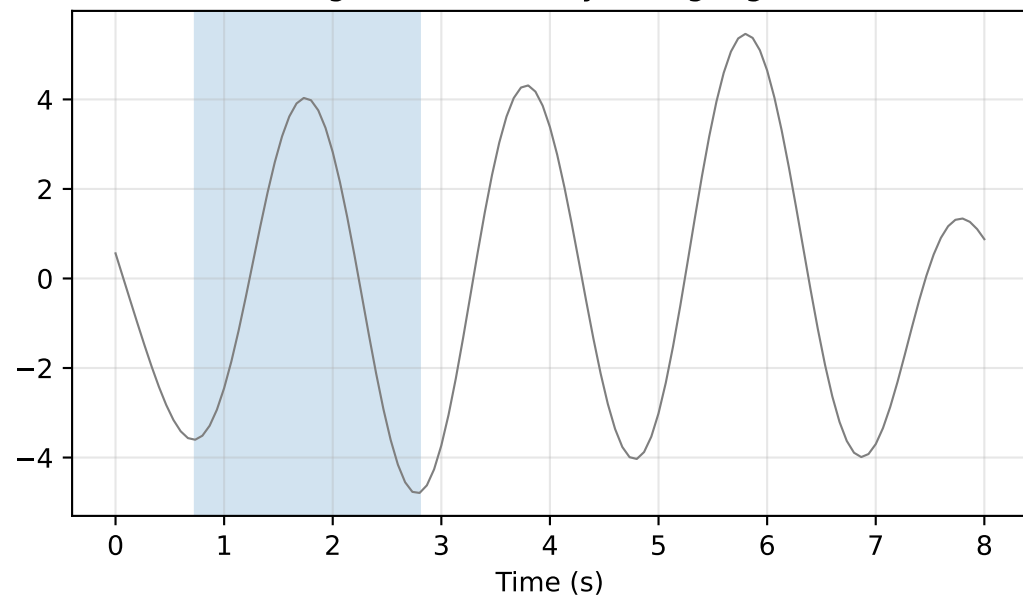


Lateral respiratory signal | $f=0.496$ Hz

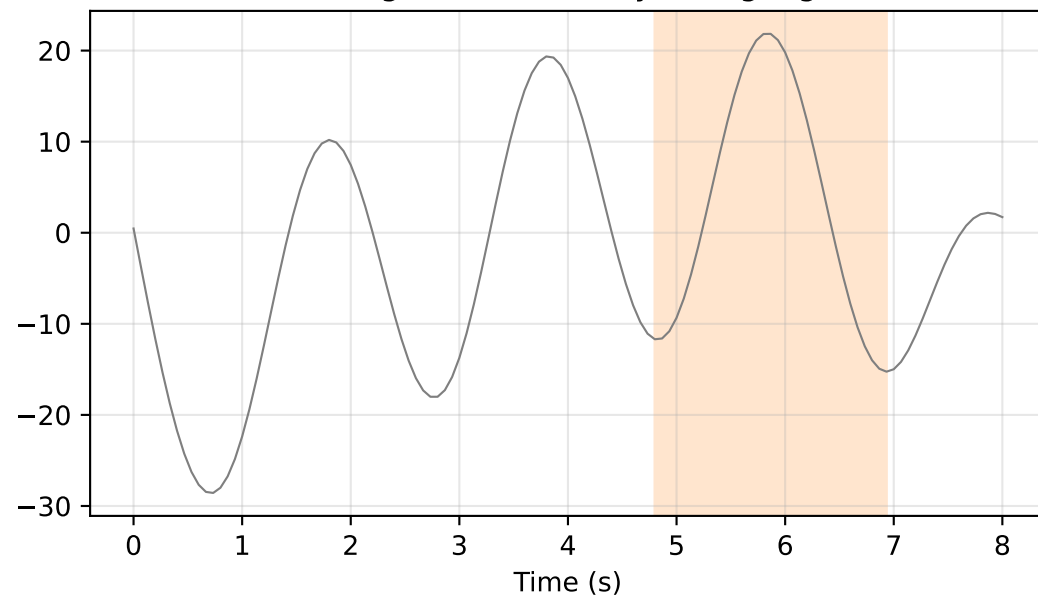


Cycle pairing

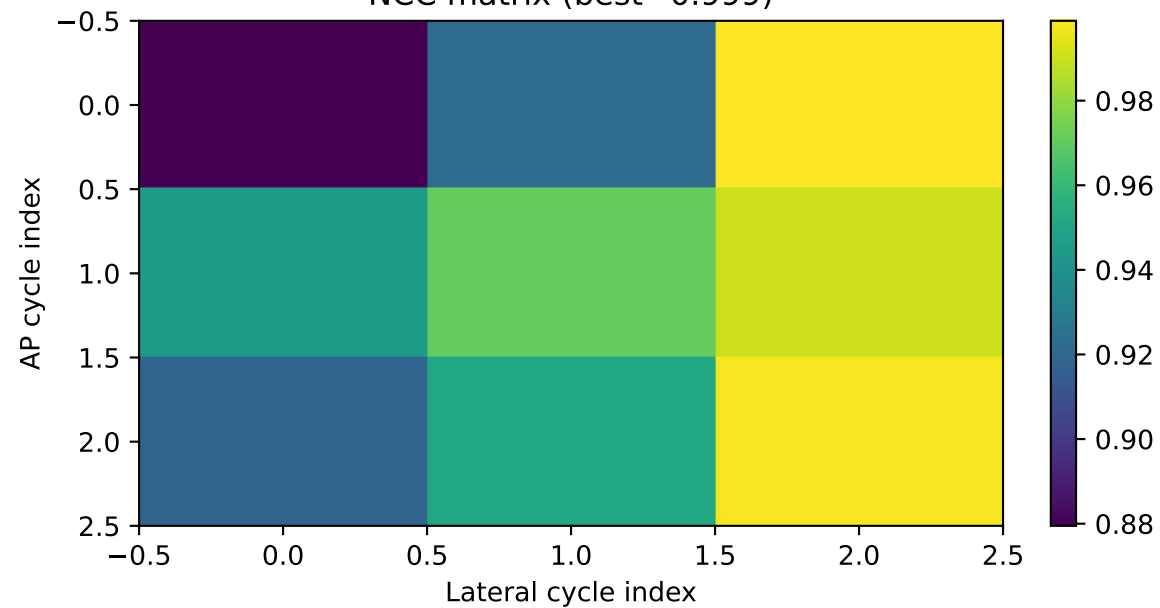
AP signal (selected cycle highlighted)



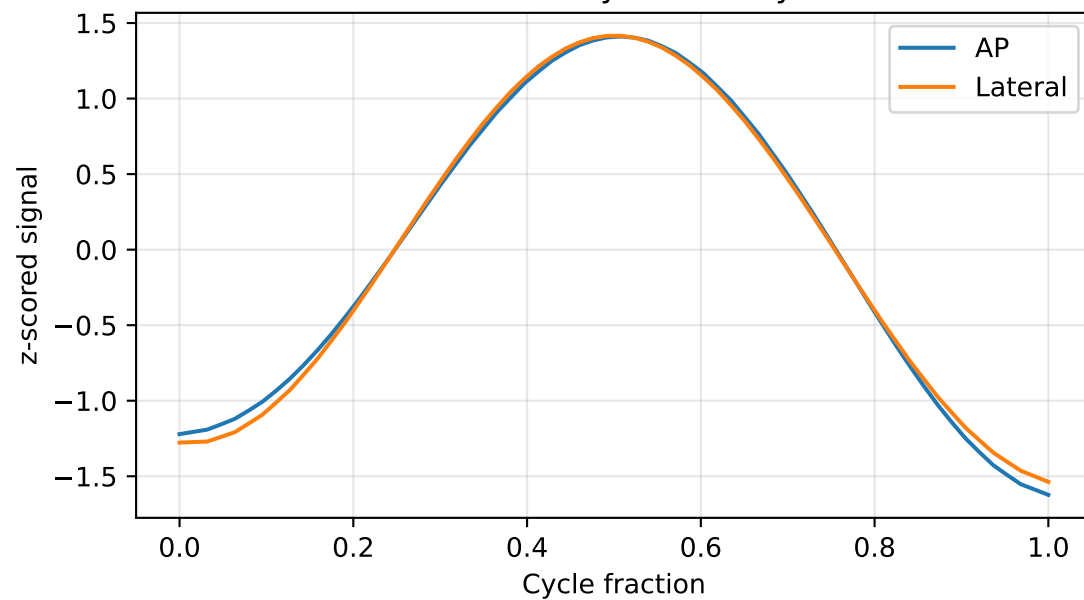
Lateral signal (selected cycle highlighted)



NCC matrix (best=0.999)



Selected cycle overlay



Metrics

Metric	Value
AP dominant frequency (Hz)	0.4959
AP period (frames)	30.25
Lateral dominant frequency (Hz)	0.4959
Lateral period (frames)	30.25
Best pair NCC	0.9989
AP cycle frames	11 - 42
Lateral cycle frames	72 - 104
min_period_s	1.0000